

LOT SYSTEMS CORPORATION

COSMO® COMPUTER – OPERATOR MANUAL

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LOT® Founded 7 April 2016 | COSMO® Founded 1 July 2024 | brand.lot-systems.com

DOCUMENT SET: LOT-COSMO-COMPUTER-v1 (product spec) + LOT-COSMO-COMPUTER-FIRMWARE + LOT-COSMO-COMPUTER-SOFTWARE

CLASS: RESTRICTED // S-2 EYES | DATE: 2026-08-27

COSMO(R) COMPUTER – PRODUCT SPEC, PLAN, BOM, ROADMAP

```
=====
LOT SYSTEMS CORPORATION
DOCUMENT: LOT-COSMO-COMPUTER-v1
TITLE:    COSMO® Computer – Physical Hardware Computer, Plan + BOM + Roadmap
CLASS:    RESTRICTED // S-2 EYES
S-2:     VADIK MARMLADOV
DATE:     2026-08-27
VERSION:  1.0 – DESIGN LOCK PENDING (PRE-PROTOTYPE)
=====
```

```
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00 // READING LOG – SOURCES THIS DOCUMENT IS BUILT ON
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```

This is a fresh v1, not a diff. `docs/benchmark/LOT-MANIFEST.md` records a prior "COSMO® Cube – complete hardware computer design v1.0" on branch `brave-lamport-t9z5u8` (14/14 iterations, BEST, +2610 lines) – that remote branch no longer exists and its content is not recoverable from this clone. This document restarts the COSMO® hardware-computer line from S-2's current brief rather than guessing at lost content, and claims the `brave-lamport` naming lineage forward from here.

Read in full before writing a line of spec:

docs/corporate/LOT_ROBOTICS_COSMO.md
COSMO® is the robotics/hardware product line named for Kuzya Cosmo Marmeladov. Section "Revenue Path" already prices Phase 3 (COSMO® Hardware, 2028-2029) at \$2,500-\$5,000/unit + \$100/mo soul-sync. This document is a smaller, nearer-term object than that phase's full robot – a pager-class desk computer, not a companion robot – but it inherits the same brand, the same "no activation without a verified LOT profile" posture, and the same father-son origin.

docs/corporate/LOT-CUBIQ-QUANTUM-CUBE-v0.md
LOT®'s CUBIQ™ is a haptic notification body – motion is its language, it has no screen, no camera, no general compute. Its own Section 00 already drew the line this document confirms: "CUBIQ™ is not that object... The two are related by lineage (father/son, LOT®/COSMO®) and should share no naming collision going forward." The COSMO® Computer specified here is that other object – a general-purpose hardware computer with a screen, a camera, and an API connector.

docs/corporate/LOT-CUBIQ-OPERATOR.md
Section 04, "AI-Driven Physical Product Delivery" and Section 07 Phase 4, "Physical Extension (Days 90+)" – the same operator-tier gating logic (behavioral verification before hardware) applies here.

docs/corporate/LOT-TERMINAL-M2M.md
The M2M data-intake protocol (device_id, operator, standardized JSON, `POST /v1/m2m/intake`) is the wire format the LOT API Connector (Section 06 below) reuses rather than reinventing.

docs/corporate/CQGS-WHITE-PAPER-SNAPSHOT.md
Confirms the Quantum Cube product line already named "bioelectric hardware, haptic feedback, nano-ceramic, piezoelectric" as CUBIQ's material register (Section II). The COSMO® Computer deliberately uses a different material register – stainless steel, not nano-ceramic – to keep the two product lines visually and physically distinct on a shelf, not just distinct in prose.

docs/corporate/LOT_QI46_ENGINE.md
Confirms the Quantum Cube as a Month-12 milestone device reporting haptic telemetry back into the Calibration Loop. The COSMO® Computer (Section 08 below) closes an analogous loop for text notifications and the "Copy" button signal, using the Logs surface already live in the product (`src/client/components/Logs.tsx`) rather than a new one.

NOTE ON SOURCES REQUESTED BUT UNREACHABLE: S-2's brief named brand.lot-systems.com, lot-systems.com/about, and institute.lot-systems.com/cqgs.html as further reading. All three are blocked by this session's network egress proxy (external fetch denied at the domain level). This document is grounded in the local corpus instead (the six documents above, all mirrors or supersets of that same brand and CQGS material already checked into this repo). Re-reading the live pages once egress is available is listed as an open item in Section 11.

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01 // WHAT THIS IS AND WHAT IT IS NOT
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```

COSMO® COMPUTER IS:

- A pocket-sized, general-purpose hardware computer: camera, screen, one button, wireless connectivity, environmental sensing, running real firmware – not a single-purpose notification toy.
- A pager for the AI age. Its primary job is to receive one short, AI-composed line of text from lot-systems.com and show it – "Coffee time!", "You haven't journaled today," "Cohort ping: 3 new" – the way a 1990s pager showed a callback number, at the speed of a glance, not a feed.
- A physical extension of an operator's existing LOT profile, gated the same way COSMO® hardware is gated everywhere else in this corpus (LOT_ROBOTICS_COSMO.md): it authenticates to one account and does nothing useful disconnected from it.

COSMO® COMPUTER IS NOT:

- CUBIQ™. No motion, no jumping, no levitation roadmap. This device sits still. Its output is text on a screen, not a body in motion.
- A phone. No app store, no general web browser, no third-party apps. One function class: receive, display, log, and – via the camera – optionally capture and relay a still frame on operator request.
- A subscription-free device. Like every COSMO® object in this corpus, it requires an active LOT profile to activate and stays useless without one.

02 // PHYSICAL FORM

DIMENSIONS	40mm x 40mm x 5mm – a flat silver square, "coin on a desk," not "gadget on a desk." The 5mm height is the hard design constraint everything else in this document is fought against: the camera module, battery, wireless-charge coil, MCU, and display all have to fit inside a stack thinner than a stack of four credit cards.
BODY	Two-piece stainless steel shell (SUS304, brushed on the internal chassis face, see finish split below), CNC-milled or precision-stamped depending on the 100-unit run's tooling economics (Section 09). Two pieces = a front face plate and a rear face plate, sandwiching the PCB stack and battery, fastened with 4x M1.4 stainless screws at the corners, gasket-sealed for light dust/splash resistance (no IP rating claimed at v1 – untested).
FACE A (FRONT)	Polished stainless steel, mirror finish. No features, no seams beyond the perimeter parting line. This face is inert by design – the device's "off" face, meant to sit face-down or face-up as a quiet object, reflecting the desk rather than competing with it.
FACE B (BACK)	Camera, screen, button – the "on" face: - CAMERA: center-top, small aperture, module detailed in Section 03. - SCREEN: center, the notification surface, detailed in Section 04. - BUTTON: bottom edge, labeled COPY, detailed in Section 05.
CHARGING	Wireless, Qi-BPP class, through Face A (the polished face doubles as the charge-contact face, the same "the charging pad IS the table" logic LOT-CUBIQ-QUANTUM-CUBE-v0.md Section 02 already established for CUBIQ – the object rests face-down to charge, face-up to be read).
MASS TARGET	<45g fully assembled. Stainless is heavy relative to plastic or nano-ceramic; the 5mm height budget forces a small-diameter coil and a thin cell, which keeps mass down despite the metal shell.

03 // CAMERA

Fixed-focus module, AI-grade off-the-shelf (Section 09 – not a hobbyist board-camera part; automotive/industrial-qualified sensor lines are specified for temperature tolerance and long-run consistency across a 100-unit batch, where part-to-part variance in a hobbyist part would show up as visible quality drift between early and late units).

RESOLUTION 2MP class (1920x1080 sufficient for its one job –

CAPTURE MODEL	an operator-triggered still capture relayed to the LOT dashboard, not video, not always-on capture). Operator-initiated only, no background/always-on capture. A short button-hold (Section 05) triggers one still frame, uploaded via the LOT API Connector (Section 06) and attached to the corresponding Log entry. This mirrors the M2M protocol's existing device_id + operator + timestamp envelope (LOT-TERMINAL-M2M.md Format 1/2) with an image payload added rather than a new schema invented.
PRIVACY POSTURE	No lens cover shipped at v1 is a known gap – see Section 11 open items. A physical shutter or status LED indicating "camera armed" is the leading candidate fix before design lock.

04 // SCREEN – THE PAGER SURFACE

CLASS	Low-power reflective/transflective LCD or e-paper class display, round or square to match the 40mm body, ~30-32mm active area. E-paper is the leading candidate: near-zero power at rest (the message persists on-screen between refreshes with no current draw), which matters directly for the battery budget inside a 5mm-thick shell.
CONTENT MODEL	One short line, AI-composed on lot-systems.com, pushed down and displayed verbatim. "Coffee time!" is the brief's own example – the design target is a single glanceable phrase, not a scrolling feed, not a dashboard. This is the same "complexity into simplicity" philosophy LOT-TERMINAL-VISION.md states directly: "Weather station with 12 sensors... OUTPUT: Air quality: 67/100... BETTER: Air quality: Good (67/100) - open your windows for 3 minutes." The COSMO® Computer screen is that BETTER line, worn on a wrist-sized object instead of read on a dashboard.
REFRESH TRIGGER	Server-pushed only. lot-systems.com decides WHEN a notification is worth a physical glance (the same AI-driven delivery-timing logic named in LOT-CUBIQ-OPERATOR.md Section 04) – the device does not poll on a fixed interval, it wakes on push.

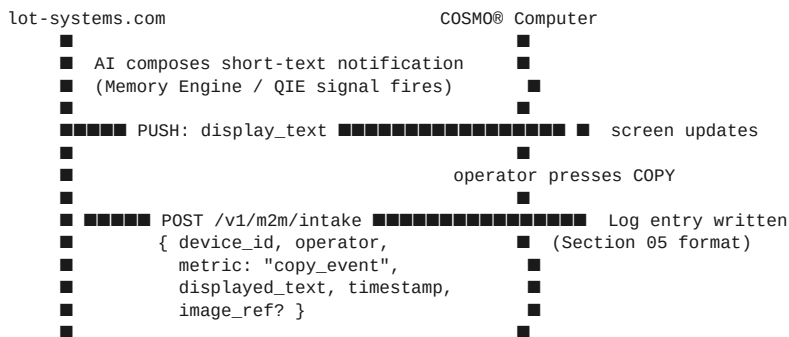
05 // THE COPY BUTTON – SIGNAL BACK TO THE LOG TAB

One physical button, Face B, labeled COPY. Two press classes:

SHORT PRESS	Sends a signal back to the operator's Log tab on lot-systems.com – the currently displayed notification is written into the account's activity log, the same log surface already live in the product (`src/client/components/Logs.tsx`). This is the hardware equivalent of hitting "save" on a passing thought: the AI said "Coffee time!", the operator glanced at it, pressed COPY, and now it exists as a timestamped Log entry rather than evaporating off the screen.
LONG PRESS (HOLD)	Triggers the camera capture path (Section 03) and attaches the resulting frame to the same Log entry.
LOG BODY FORMAT	Per the repo's own COCKPIT-RULE convention (docs/benchmark/LOT-LEXICON.md: "Log body = instrument readings only; label names the event; no narration"), the Log entry this button writes is data, not prose: label: "COSMO-COMPUTER: COPY" body: device_id, operator, displayed_text, timestamp, (optional) image_ref No narration is generated by the device. The AI narrative, if any, is composed server-side from this raw entry the same way every other Log entry in the product already is.

06 // LOT API CONNECTOR

The device speaks the existing M2M data-intake protocol (docs/corporate/LOT-TERMINAL-M2M.md) rather than a new one:



TRANSPORT TLS 1.3+, matching LOT-TERMINAL-M2M.md's existing "Data Standards" clause. Device-side, a low-power WiFi or BLE-to-hub radio (Section 09 BOM) carries the connection; the exact radio choice is a prototype-stage decision, not locked at v1.

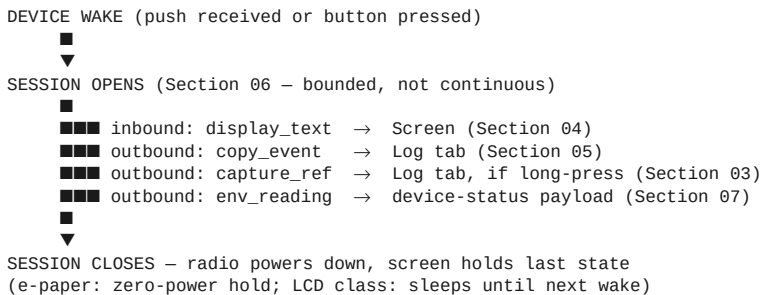
AUTHENTICATION Operator token, provisioned once during setup (paired via the companion software, see docs/technical/LOT-COSMO-COMPUTER-SOFTWARE.md), matching the existing `Authorization: Bearer <operator\_token>` scheme already documented for the M2M intake endpoint.

SESSION COMPRESSION Per S-2's brief ("compress the information in each session"): the device does not stream continuously. Each wake-push-log cycle is treated as one bounded session – connect, receive or send one payload, disconnect. Firmware buffers any queued Log events (e.g. COPY presses while offline) and flushes them as a single batched payload on next connect, rather than holding a live connection open. This is conceptually the same instinct behind the product's existing Memory Engine compression work (docs/technical/MEMORY-ENGINE-COMPRESSION-ARCHITECTURE.md) applied to a battery-constrained device: say the necessary thing once, then go quiet.

07 // WEATHER SENSOR

One onboard environmental sensor package – temperature, relative humidity, barometric pressure – AI-grade off-the-shelf (Section 09), the same class of part already named in the product's own hardware precedent (docs/technical/LOT-NODE-0-RIG-SPEC.md's "AI-grade" sourcing standard applied at rig scale; here at sensor scale). Readings ride the same LOT API Connector session model as Section 06, folded into the periodic device-status payload rather than a separate always-on stream – this device is a pager first, a weather station second. Sensor fusion with the wider LOT-TERMINAL-M2M.md "Multi-Sensor Array" format (Format 3) is the natural home for this data once the device ships.

08 // DATA MODEL – HOW A SESSION BECOMES A LOG



This is the "session compression" S-2's brief names directly: nothing about this device holds a persistent connection or streams continuously. Every session is intake → classify → action → record, at device scale – the identical four-step discipline already governing the LOT server side in docs/technical/LOT-NODE-0-RIG-SPEC.md Section 04 ("INPUT → CLASSIFY → ACTION → RECORD").

Prices are current street-range ESTIMATES for a 100-unit prototype/pilot run, not confirmed quotes – the same disclosure standard docs/technical/LOT-NODE-0-RIG-SPEC.md uses for its own BOM ("current street ranges... both move week to week"). Real quotes are a Section 10 action item before any PO is cut. Vendor names are real, well-known component/manufacturing suppliers; no specific SKU or deep product-page link is asserted here, since none has been verified against current availability.

...

COMPONENT SPEC / CLASS VENDOR CLASS EST. UNIT COST

PCB fab + assembly	4-layer rigid, <=40mm x 40mm outline (pcbway.com) – SMT assembly + fab in one order, matched to a 100-unit pilot MOQ	PCBWay	\$8 - \$18
MCU / SoC	Low-power ARM Cortex-M/M-class SoC with integrated BLE or WiFi radio (final part TBD at prototype stage)	Digi-Key / Mouser (digikey.com, mouser.com)	\$3 - \$8
Camera module	2MP class, fixed-focus, automotive/ industrial-qualified sensor line (AI-grade – Section 09a)	Digi-Key / Mouser / specialty camera-module distributors	\$4 - \$9
Display	E-paper or transfective LCD, ~30-32mm active area, ultra-low standby power	Digi-Key / Mouser / Good Display / Adafruit (adafruit.com)	\$6 - \$14
Weather sensor	Combined temp/humidity/pressure, industrial-grade package (Section 09a)	Digi-Key / Mouser	\$2 - \$5
Wireless charge coil	Qi-BPP receiver coil, low-profile (<=1mm), matched to 5mm body height	Digi-Key / Mouser / Würth Elektronik	\$2 - \$4
Battery	Thin-profile LiPo, capacity sized to fit remaining Z-height after coil + PCB stack (exact mAh: prototype-stage decision, Section 10)	Digi-Key / cell supplier direct	\$2 - \$5
Button (COPY)	Low-profile tactile switch, IP-rated cap, panel-mount through Face B	Digi-Key / Mouser	\$0.50 - \$1.50
Stainless shell (2 pieces)	2-piece SUS304, one CNC or stamped per piece, one face mirror-polished, one face machined for camera/screen/ button cutouts + gasket channel	PCBWay (metal fabrication service, same vendor as PCB fab) or a dedicated metal-shop partner	\$10 - \$25
Fasteners + gasket	4x M1.4 stainless screws, silicone gasket ring	McMaster-Carr / PCBWay hardware add-on	\$0.50 - \$1.50
EST. TOTAL PER UNIT (components + shell, excl. NRE)			
100-UNIT RUN, COMPONENTS + SHELL ONLY			≈ \$38 - \$91
			≈ \$3,800 - \$9,100

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09a – "AI-GRADE OFF-THE-SHELF" MEANS

Per S-2's brief item 15: industrial- or automotive-qualified part grades (extended temperature range, tighter manufacturing tolerance, longer datasheet-guaranteed lifetime) rather than consumer/hobbyist grades, for the camera and weather sensor specifically. This is a sourcing standard, not a claim that any part itself runs AI – it means "graded for unattended, long-run reliability," the same standard docs/technical/LOT-NODE-0-RIG-SPEC.md applies to ECC memory on an always-on server, applied here to sensors on an always-carried device.

NOT INCLUDED ABOVE (separate NRE / one-time costs)

- PCBWay tooling / stamping die setup for the stainless shell (one-time, amortized across the 100-unit run – real figure requires a PCBWay/metal-shop quote, Section 10).
- Firmware development, companion-software development (labor, not BOM – see the two companion documents, Section 12).
- Certification (FCC/CE for the radio; not scoped at v1 – Section 11).
- Packaging + the PDF manual print run, if physical copies are produced (Section 12 ships a digital PDF regardless).

v0.1 – PAPER DESIGN (THIS DOCUMENT)

Plan, BOM estimate, physical form, data model. No hardware exists yet. GATE: S-2 review + design lock on dimensions, face layout, and the two-document firmware/software split (Section 12).

v0.2 – BREADBOARD PROTOTYPE

Off-the-shelf dev boards (not the final 40x40x5mm shell) proving the four hard subsystems independently: (a) e-paper or LCD refresh from a server push, (b) camera capture + upload on button hold, (c) Qi charge-coil power budget inside the target battery capacity, (d) the bounded session model (Section 08) against the live M2M intake endpoint. GATE: all four subsystems independently working against real lot-systems.com infrastructure, even if not yet in one shell.

v0.3 – FIRST FITTED PROTOTYPE

Custom PCB (PCBWay fab, Section 09) fitted inside a 3D-printed or machined mockup shell at the true 40x40x5mm dimension – proves the Z-height budget actually closes with real components, not dev-board approximations. GATE: full assembly fits, powers on, completes one real end-to-end session (push → screen → COPY → Log tab entry) inside the target shell dimension.

v1.0 – STAINLESS PILOT UNIT

Final two-piece stainless shell (Section 02), both finishes (polished Face A, machined Face B), first units off PCBWay's metal fabrication line. GATE: 10 units built, 10/10 pass a basic reliability pass (charge cycle, camera capture, screen refresh, COPY-to-Log round trip) before committing to the full run.

PILOT – 100-UNIT RUN

Full BOM (Section 09) ordered at 100-unit MOQ once v1.0's 10-unit gate is clean. GATE: PCBWay quote confirmed in writing, firmware frozen (docs/technical/LOT-COSMO-COMPUTER-FIRMWARE.md marked RELEASE, not DRAFT), companion software's provisioning flow tested against at least one real operator account end-to-end.

11 // OPEN ITEMS (do not silently resolve – decide, then update this doc)

- Camera privacy: no physical shutter or "armed" indicator specified yet (Section 03). Needs a decision before v0.3.
- Radio choice (BLE-to-hub vs. onboard WiFi) not locked (Section 06) – directly affects battery budget and therefore the Z-height stack.
- No IP/water/dust rating targeted at v1 (Section 02) – decide whether the pilot run needs one before committing to a gasket spec.
- Regulatory certification (FCC/CE for the radio) not scoped anywhere in this document – required before any unit ships to an operator outside a controlled pilot.
- brand.lot-systems.com, lot-systems.com/about, and institute.lot-systems.com/cqgs.html were requested reading sources that this session's network egress blocked (Section 00). Re-check against the live pages once reachable, in case brand or CQGS material has moved since the local snapshots this document relied on.
- Real PCBWay quotes (PCB + stainless shell) not yet obtained – Section 09's BOM is estimate-only.

12 // COMPANION DOCUMENTS

Per S-2's brief (items 9-11: firmware documents, software to connect with firmware, kept as separate documents):

docs/technical/LOT-COSMO-COMPUTER-FIRMWARE.md

On-device firmware: MCU architecture, driver stack (display, camera, weather sensor, wireless charge management, COPY button), the session state machine (Section 08), OTA update path, and the LOT API Connector wire implementation (Section 06).

docs/technical/LOT-COSMO-COMPUTER-SOFTWARE.md

Server + companion-app side: the lot-systems.com endpoint that composes and pushes the pager text, the provisioning flow that pairs a physical unit to one operator token, and the Logs-tab write path the COPY button drives.

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COSMO(R) COMPUTER – FIRMWARE SPECIFICATION

```
=====
LOT SYSTEMS CORPORATION
DOCUMENT: LOT-COSMO-COMPUTER-FIRMWARE
TITLE:    COSMO® Computer – Firmware Specification
CLASS:    RESTRICTED // S-2 EYES
S-2:      VADIK MARMELODOV
DATE:     2026-08-27
VERSION:   0.1 – DRAFT (companion to docs/corporate/LOT-COSMO-COMPUTER-v1.md)
=====
```

This is the firmware half of the two-document split S-2's brief requires (item 11: firmware documents and connecting software kept separate). Read `docs/corporate/LOT-COSMO-COMPUTER-v1.md` first – it defines the physical device, the session model, and the wire protocol this firmware implements. The companion `LOT-COSMO-COMPUTER-SOFTWARE.md` covers the server and provisioning side.

01 // TARGET PLATFORM

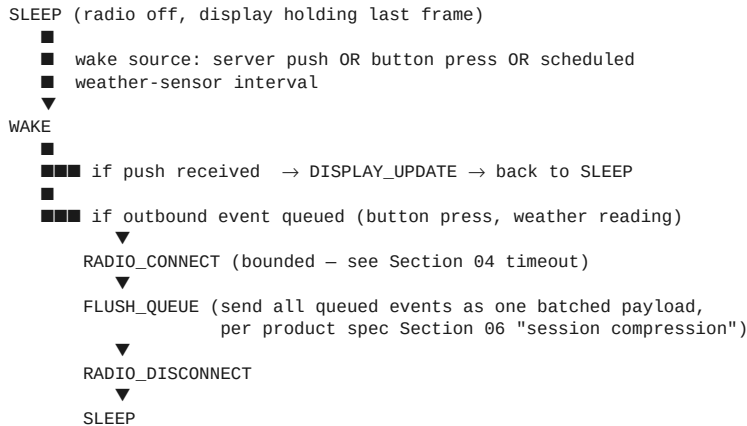
MCU CLASS	Low-power ARM Cortex-M-class SoC with an integrated low-power radio (BLE or WiFi – Section 05 of the product spec leaves the final radio choice open; firmware targets both classes behind one radio abstraction so the choice can be made at the v0.2 breadboard stage without a firmware rewrite).
RUNTIME	Bare-metal or a minimal RTOS (FreeRTOS-class), decided at prototype stage by whichever the chosen display + camera driver stack most naturally supports. No general OS, no filesystem beyond a small config/credential store – this device runs one application, not a platform.
MEMORY FOOTPRINT	Sized to the MCU's on-chip flash/RAM; no external storage is planned at v1 (nothing on-device needs to persist beyond the current display buffer, the operator token, and a small queued-event buffer for offline COPY presses – Section 04).

02 // DRIVER STACK

DISPLAY	Single-line text render to the e-paper or transreflective LCD (product spec Section 04). Driver responsibility: accept a UTF-8 string over the session payload, render it centered on the ~30-32mm active area, hold the frame with zero (e-paper) or minimal (LCD sleep mode) standby current until the next push.
CAMERA	Single-shot capture on trigger (long COPY press), JPEG or equivalent compressed encode on-device before upload – the device never buffers or transmits raw sensor frames, to keep the queued-event payload (Section 04) small on a battery-constrained radio budget.
WEATHER SENSOR	Polled once per session wake (product spec Section 07), not continuously – a continuously-polled sensor on an MCU with no external storage would either be discarded between sessions or force a persistent connection this device deliberately avoids (Section 06 of the product spec, "session compression").
WIRELESS CHARGE MGMT	Qi-BPP receiver IC handles the analog charge path; firmware's role is limited to reading charge state (charging / charged / not-present) for status reporting and to gating radio wake behavior – the device does not attempt to transmit while actively drawing peak charge current, to keep the instantaneous power budget inside the coil's rated output.
BUTTON (COPY)	Debounce GPIO interrupt, two press classes (short / long-hold) per product spec Section 05. Short press queues a copy_event; long press additionally triggers the camera driver before queuing.

03 // SESSION STATE MACHINE

Implements product spec Section 08 directly:



No state holds the radio connected without an active reason. This is the firmware-level enforcement of the product spec's core constraint: every session is bounded, nothing streams continuously.

04 // OFFLINE QUEUE + CONNECTION BUDGET

QUEUE A small fixed-depth ring buffer of pending outbound events (copy_event, capture_ref, env_reading). Sized to hold a full day's worth of plausible button presses (single digits – this is a pager, not a logger) plus one queued image reference. Overflow policy: drop oldest non-image event first – an image reference is treated as higher-value than a redundant status ping.

CONNECTION TIMEOUT RADIO_CONNECT (Section 03) is bounded by a hard timeout (exact value: prototype-stage tuning, target order-of-magnitude: single-digit seconds for a successful handshake). A timeout with events still queued returns to SLEEP and retries on the next wake rather than looping the radio in a wait state – a device this battery-constrained cannot afford to spin on a bad connection.

05 // WIRE PROTOCOL – LOT API CONNECTOR

Implements product spec Section 06, reusing the existing M2M intake envelope (``docs/corporate/LOT-TERMINAL-M2M.md``) rather than a bespoke schema:

```
...
Outbound (device -> lot-systems.com), one call per session flush:

POST /v1/m2m/intake
Authorization: Bearer <operator_token>
Content-Type: application/json

{
  "device_id": "cosmo-computer-<serial>",
  "operator": "<operator_id>",
  "events": [
    { "metric": "copy_event", "displayed_text": "...",
      "timestamp": "...", "image_ref": "..." },
    { "metric": "env_reading", "temperature_c": ..., "humidity_pct": ...,
      "pressure_hpa": ..., "timestamp": "..." }
  ]
}
...

Inbound (lot-systems.com -> device), server-initiated push:

{ "display_text": "Coffee time!", "timestamp": "..." }
```

...

BATCHING NOTE The base M2M protocol (LOT-TERMINAL-M2M.md) shows one-metric-per-call examples (Format 1/2/3). This firmware wraps outbound calls in an `events` array specifically to implement the session-compression requirement (product spec Section 06) – multiple queued events flush as one call, not one call per event. This is a firmware-side batching convention on top of the existing envelope, not a protocol version change; the server-side handling of this shape is specified in the companion software document.

TRANSPORT TLS 1.3+, matching LOT-TERMINAL-M2M.md's Data Standards clause.

IMAGE PAYLOAD `image\_ref` in the outbound events array is resolved to actual image bytes via a follow-up multipart or presigned-upload step – the exact transport is a companion-software decision (see `LOT-COSMO-COMPUTER-SOFTWARE.md`), kept out of this firmware document because it depends on server-side storage choices firmware should not need to know about.

06 // OTA UPDATE PATH

Not scoped for v0.1-v0.3 prototypes. Required before the 100-unit pilot run (product spec Section 10, PILOT gate: "firmware frozen... marked RELEASE, not DRAFT") since a fielded batch of 100 units with no update path means any post-ship bug is permanent per-unit. Candidate approach: signed-firmware-image pull over the same session-bounded connection model (Section 03), triggered by an explicit server flag rather than polled – consistent with the device's "wake only when there's a reason" design. Left as an open item for the v0.3 -> v1.0 transition, not designed here.

07 // SECURITY POSTURE

- Operator token stored in on-chip secure storage (if the chosen MCU provides it) or, at minimum, not stored in plaintext in application flash. Exact mechanism depends on the final MCU selection (Section 01).
- No device ever holds more than one operator token at a time – re-pairing (companion software) explicitly overwrites, never appends.
- Camera trigger is always operator-initiated (long COPY press); there is no remote or scheduled capture path in this firmware. This is a firmware-level guarantee, not just a policy statement – the capture driver (Section 02) has no code path that fires without the button interrupt.

=====
AUTHORIZED BY: S-2 // VADIK MARMELADOV
END LOT-COSMO-COMPUTER-FIRMWARE
=====

COSMO(R) COMPUTER – SERVER + COMPANION SOFTWARE SPECIFICATION

```
=====
LOT SYSTEMS CORPORATION
DOCUMENT: LOT-COSMO-COMPUTER-SOFTWARE
TITLE:    COSMO® Computer – Server + Companion Software Specification
CLASS:    RESTRICTED // S-2 EYES
S-2:      VADIK MARCELADOV
DATE:      2026-08-27
VERSION:   0.1 – DRAFT (companion to docs/corporate/LOT-COSMO-COMPUTER-v1.md)
=====
```

This is the software half of the two-document split S-2's brief requires (item 11: firmware and connecting software kept separate). Read `docs/corporate/LOT-COSMO-COMPUTER-v1.md` first for the device and wire protocol, and `docs/technical/LOT-COSMO-COMPUTER-FIRMWARE.md` for exactly what the device sends and expects. This document covers three things: the lot-systems.com endpoint that composes and pushes pager text, the provisioning flow that pairs one physical unit to one operator, and the server-side handling of what the firmware sends back.

01 // WHERE THIS SITS IN THE EXISTING STACK

No new backend framework – this rides the same Fastify + PostgreSQL stack already running lot-systems.com (`src/server/`). Three additions:

1. An outbound push path: something in the existing signal-routing substrate (QIE – Quantum Intention Engine, see LOT-LEXICON.md) decides a short line of text is worth sending to a paired device, and a delivery service turns that decision into the firmware's inbound `{ "display\_text": ... }` payload (product spec Section 06).
2. An inbound intake path: the existing M2M intake route (`POST /v1/m2m/intake`, docs/corporate/LOT-TERMINAL-M2M.md) gains handling for the `events` array batching shape the firmware sends (LOT-COSMO-COMPUTER-FIRMWARE.md Section 05) – an extension of the existing route's payload shape, not a new route.
3. A provisioning flow (Section 03 below) that does not exist yet anywhere in the M2M protocol, since the S-2 Operator model in LOT-TERMINAL-M2M.md pairs a *maker's* hardware to their own account; COSMO® Computer pairing is closer to "activate a purchased consumer device," which needs its own short flow.

02 // NOTIFICATION COMPOSITION + PUSH

SOURCE OF TEXT	Composed by the same AI layer already generating Memory Engine questions and Log narration elsewhere in the product – not a new model, a new delivery surface for existing composition capability. Input: one QIE signal (badge unlock, memory question ready, cohort ping, a scheduled-job output like the brief's own "Coffee time!" example). Output: one short line, length-bounded to what the device's ~30-32mm single-line screen (product spec Section 04) can render without truncation-mid-word.
DELIVERY DECISION	Not every QIE signal becomes a push – the same "AI decides WHAT/WHEN/HOW to send" logic named in docs/corporate/LOT-CUBIQ-OPERATOR.md Section 04 governs this device too. A rate limit (exact cadence: operator-configurable, default TBD at prototype stage) prevents the device from becoming a second feed – the whole point of a pager-class object is that it says less, not more, than a phone.
PUSH TRANSPORT	Server holds no persistent connection to the device (firmware Section 03 – device is asleep by default). Delivery is queued server-side and delivered on the device's next scheduled wake or via a low-power wake mechanism appropriate to the final radio choice (BLE: connection-oriented wake; WiFi: a lightweight push service) – the exact wake-trigger mechanism is a v0.2 breadboard-stage decision (product spec Section 10), not fixed here.

03 // PROVISIONING – PAIRING ONE UNIT TO ONE OPERATOR

1. Operator receives a physical unit (out of the 100-unit pilot run,

- product spec Section 10) with a unique device_id printed or encoded (e.g. QR on the packaging, not on the polished shell itself – product spec Section 02 keeps Face A feature-free).
2. Operator scans/enters device_id on lot-systems.com, authenticated into their existing LOT profile.
 3. Server checks Benchmark/Usership eligibility per the same gating posture named in docs/corporate/LOT_ROBOTICS_COSMO.md ("A COSMO® unit without a verified LOT profile does not activate") – the eligibility bar for a pager-class device is a policy decision left open here, not assumed to be the same Purple-tier bar set for full COSMO® robotics hardware in that document.
 4. Server mints a scoped operator_token for that device_id and holds it pending device-side confirmation.
 5. Device, powered on for the first time (or after a factory reset), enters a pairing-wait state and receives the token on first successful radio handshake, storing it per firmware Section 07.
 6. Server marks the device_id ACTIVE, tied to the operator's account. Re-provisioning (device changes hands, or is reset) explicitly revokes the prior token before issuing a new one – never two live tokens for one device_id.

04 // HANDLING THE FIRMWARE'S BATCHED EVENTS

Server-side handling of the `events` array the firmware sends (LOT-COSMO-COMPUTER-FIRMWARE.md Section 05):

copy_event	Written to the operator's Log tab (`src/client/components/Logs.tsx` surfaces this to the operator) using the COCKPIT-RULE format the product spec's Section 05 already defines: label names the event ("COSMO-COMPUTER: COPY"), body carries the raw fields (device_id, operator, displayed_text, timestamp, image_ref), no server-generated narration is injected into the Log body itself – any narrative framing happens at display time in the Logs UI, same as every other Log entry class in the product today.
env_reading	Folded into the device's status record, and made available to the same M2M "Multi-Sensor Array" surface (LOT-TERMINAL-M2M.md Format 3) other hardware in the ecosystem already uses – no bespoke weather-data table.
image_ref	Resolved against whatever object-storage path the device's follow-up upload used (firmware Section 05 leaves the exact transport to this document – the concrete choice, e.g. presigned upload URL returned in the same intake response, is a prototype-stage decision, not fixed here). Attached to the copy_event's Log entry as image_ref once resolved.

05 // PDF MANUAL GENERATION

Per product spec Section 12 and S-2's brief item 7 ("result in PDF manuals"): the operator-facing manual is generated from this documentation set (product spec + this document + the firmware document), not hand-maintained separately, the same "generate the PDF from the markdown source of truth" pattern already used for
`docs/badges/LOT\_BADGES\_ACHIEVEMENTS\_MASTER\_CODEX\_v\*.md` ->
`.pdf` in this repo (see `scripts/generate\_badge\_pdf\_v31.py` and siblings for the existing tooling pattern this reuses). A COSMO® Computer manual PDF is produced this session (Section 06) as the first instance of that pipeline applied to this product line.

06 // WHAT IS EXPLICITLY NOT DESIGNED HERE

- The exact low-power wake-trigger mechanism (Section 02) – depends on a radio choice not yet locked (product spec Section 11 open items).
- Rate-limit defaults for push cadence (Section 02) – an operator-facing settings decision, not a hardcoded constant.
- Object storage choice for image_ref resolution (Section 04) – an infrastructure decision that should reuse whatever the product already uses for other user-uploaded media, not invent a new store.
- Benchmark/Usership eligibility tier required to provision a unit (Section 03, step 3) – a business decision for S-2, not a technical one, and explicitly left unset rather than guessed.

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END LOT-COSMO-COMPUTER-SOFTWARE

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COSMO® COMPUTER – v1.0 DESIGN LOCK PENDING (PRE-PROTOTYPE)
100-UNIT PILOT RUN | PCBWay FABRICATION | LOT API CONNECTOR (M2M PROTOCOL)
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